

Markscheme

Mathematics: analysis and approaches
Practice question bank

89. (a)

The sum of any three consecutive integers can be written as $n + (n + 1) + (n + 2) = 3n + 3 = 3(n + 1)$, which proves the sum is always a multiple of 3.

Using this result, find the value of k such that $14 + 15 + 16 = 3k$.

$$14 + 15 + 16 = 45 \qquad (M1)$$

$$45 = 3(15), \text{ so } k = 15 \qquad \mathbf{A1}$$

[2 marks]